

SHIRSENDU TALUKDER

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PROFESSIONAL SUMMARY

Mathematics postgraduate with CSIR–NET LS qualification (AIR 31) and GATE Mathematics qualification (AIR 98). Research interests include Commutative Algebra, Category Theory, Algebraic Geometry. Experienced in mathematical exposition, LaTeX typesetting, Python-based visualization, and computational mathematics.

EDUCATION

M.Sc. in Mathematics

2023–2025

Ramakrishna Mission Vivekananda Educational and Research Institute, Howrah, West Bengal CGPA: 7.88/10

B.Sc. in Mathematics

2020–2023

The Heritage College, University of Calcutta CGPA: 8.367/10

ACHIEVEMENTS

CSIR–NET LS (Mathematical Sciences), June 2025, AIR: 31

GATE Mathematics (MA), 2026, Score: 638, AIR: 98

RESEARCH

M.Sc. Thesis: *Homological Methods in Commutative Algebra*

2025

Supervisor: Dr. Sagnik Chakraborty [\[View Thesis\]](#)

Natural Transformations: A Study Through Exercises, Proofs and Visual Interpretation

2025

[\[View Notes\]](#)

PROJECTS

- ▶ **DesmosImageRenderer** — implemented pixel plotting and linear transformation visualization using Python (OpenCV, PIL, NumPy), Flask, and the Desmos API
- ▶ **Turtle-Image-Renderer** — Python application that reconstructs images using Turtle Graphics.
- ▶ **Processing Simulations** — Implemented mathematical and physical simulations including the Double Pendulum and Perlin-noise-based visual systems.
- ▶ **TouchDesigner Visualizations** — Created procedural animations and real-time interactive graphics using node-based visual programming.

SKILLS & COURSEWORK

Technical Skills:

Python • LaTeX • OpenCV • Processing • Manim • TouchDesigner

Relevant Coursework:

Commutative Algebra • Category Theory • Algebraic Geometry • Algebraic Topology • Lie Algebra • Functional Analysis • Measure Theory • Field Theory